G · Standing the tops and fitting the rings

Two people. Both tops lie face down with their legs, feet and rails on (chapter F), every OLOV at 80.2 cm and every pad set with the foot height gauge (102 mm). ALEX 1 stands in its place; ALEX 2 stands 5 cm short of its place, its near end 282.5 cm from the back wall. Neither has rings yet: each ring is placed by tracing round its pad.

Parts	Hardware	Tools
P01 top A · P02 top B · P03 ALEX 1 · P04 ALEX 2 · P05 OLOV leg · P06 rail A · P07 rail B · P09 foot · P10 ring × 8	H06 #6 × 5/8 in flat head × 24 (3 per ring)	Sharp pencil, wall gauge, ALEX corner template, drill guide (SET tower, 12 ALEX puck), 5/64 in bit, level, tape measure, blanket

1 **Stand top A on the bare ALEX 1.** With a helper, turn top A over and stand it on its legs with its four pads on the bare ALEX top: lower the ALEX end first, then the leg end. Push the ALEX unit, never the top, until 3.5 cm of top shows all round the cabinet and the wall gauge's ALEX 65 leg (6.5 cm) just fits between the back wall and the ALEX.

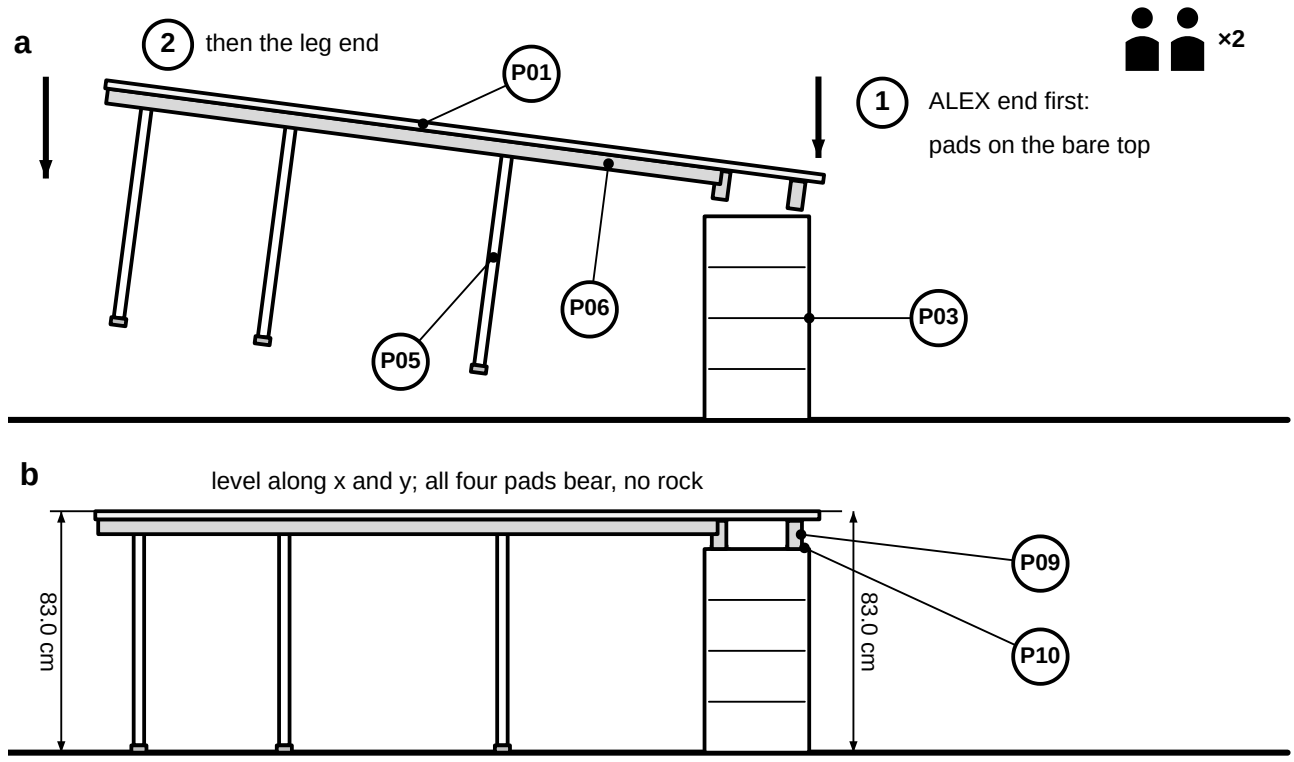


Fig. G-1 · Top A seen from the room, looking at the back wall; x grows to the right (1:26). a: step 1, ALEX end first, the pads land on the bare ALEX top; b: step 3, back on its legs with the pads in their rings, level, 83.0 cm at both ends. Rail A hangs under the top; it never touches ALEX 1.

! **Push the cabinet, never the top.**
Top A is positioned only by moving ALEX 1 under it. Once the rings are on and the pads sit in them, push the ALEX with the pads in the rings; the top moves with it.

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G · Standing the tops and fitting the rings 1

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J · Wall-gap trim (optional) 13

2

Fit the rings on ALEX 1 by tracing.

1. With top A standing where step 1 left it, draw round each of the four pads on the ALEX top with a sharp pencil.
2. Two people: lift top A off and lay it face down on the blanket.
3. **Check the traces before drilling anything.** Lay the ALEX corner template on each bare corner, fences against the edges: the pencil trace must sit inside the template's inner engraved circle (⌀52 mm) all round. If one does not, put the top back on, push the cabinet to correct it, and trace again.
4. Set a ring (P10) on each pencil circle, flange down. Look down through its open bottom and move it until the pencil line shows all the way round inside the ⌀52 mm bore.
5. Hold it there. Start the three 5/64 in pilots through the ring's own holes, 3–4 mm deep, lift the ring, and finish each pilot 12 mm deep with the drill guide's **12 ALEX** puck.
6. Put the ring back and drive three H06 #6 × 5/8 in flat-head screws.

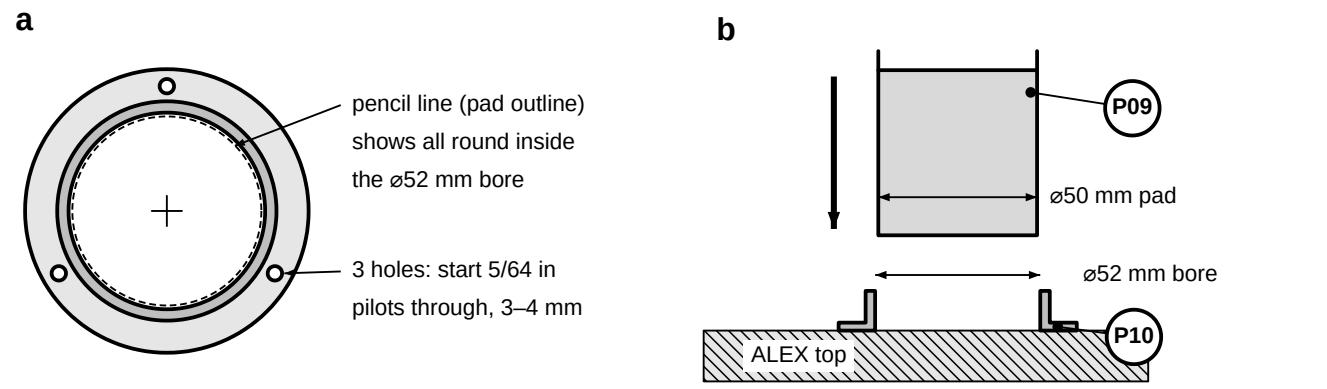


Fig. G-2 · P09 foot, P10 ring. a: a ring on its pencil circle, seen from above (1:2); b: the pad drops into the ring, section (1:2.4). Ring bore 52 mm VERIFIED (from its geometry file); pad 50 mm MEASURE ON HARDWARE. The corner-template check (item 3) is made on the bare corner, before any ring is fitted: a flat template cannot lie over a fitted ring.

3

Stand top A again and level it. Two people, ALEX end first as in step 1: the pads drop into the rings. Level along x and along y by turning the OLOV feet and the round feet until it reads flat and all four pads bear with no rock. Target: surface 83.0 cm above the floor at every corner. SOURCE-DEFINED

!

From here on, never push or lift the top alone.
If top A is not 3 cm off a wall (the wall gauge's EKB 30 leg, 3 cm), push the ALEX with the pads in the rings; the top moves with it.

83.0 cm assumes ALEX units 70.0 cm tall ASSUMPTION / FIELD CHECK until measured. With the optional ALEX feet the surface is 85.0, 87.0 or 89.0 cm (2, 4 or 6 cm feet).

4

Top B on the bare ALEX 2, then its rings.

1. Wind each of top B's OLOV legs to 79.9–80.0 cm (plate face to foot), so rail B's corner end can later slide under top A.
2. ALEX 2 stands with its near end 282.5 cm from the back wall (chapter F), 5 cm short of its place, so rail B's corner end clears rail A's flat leg. Turn top B over onto its legs with its pads on the bare ALEX 2, ALEX end first. For a moment top B's far end reaches 319 cm from the back wall: the side wall must run at least 322 cm; if it does not, stop and ask.
3. Push ALEX 2, never the top, until 3.5 cm of top shows all round the cabinet and the wall gauge's ALEX 65 leg (6.5 cm) just fits between the side wall and the ALEX.
4. Fit the four rings on ALEX 2 by tracing, exactly as in step 2: pencil round each pad, lift top B off, check every trace with the corner template, rings on the circles, pilots through the rings, screws.
5. Stand top B back on, ALEX end first: the pads drop into the rings.

5

Push ALEX 2 home; top B comes with it. Push ALEX 2 the last 5 cm toward the back wall, with the pads in its rings, until rail B's corner end touches the outer face of rail A's vertical leg and B's end meets A's front edge (y = 65 cm). Under the corner, rail B's outer face lines up with the taped 52.0 cm mark on rail A. Then wind B's legs up to 80.2 cm, until the two surfaces are flush.

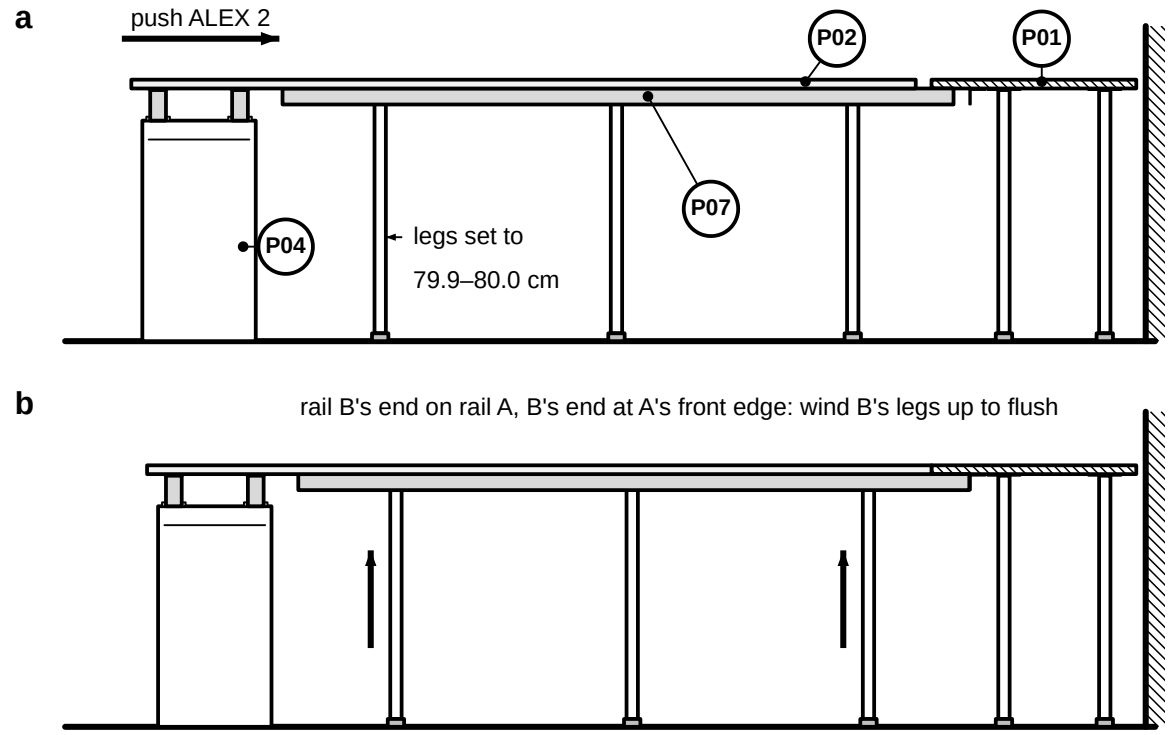


Fig. G-3 · P01 top A, P02 top B, P04 ALEX 2, P07 rail B. Seen from the knee space, looking at the side wall: top A is on your right, y grows to the left (1:24); top A cut beyond B's inside edge, ALEX 1 not shown. a: rings on, legs at 79.9–80.0 cm, ALEX 2 5 cm out, push it home; b: home, legs wound up to 80.2 cm. Neither rail touches an ALEX.

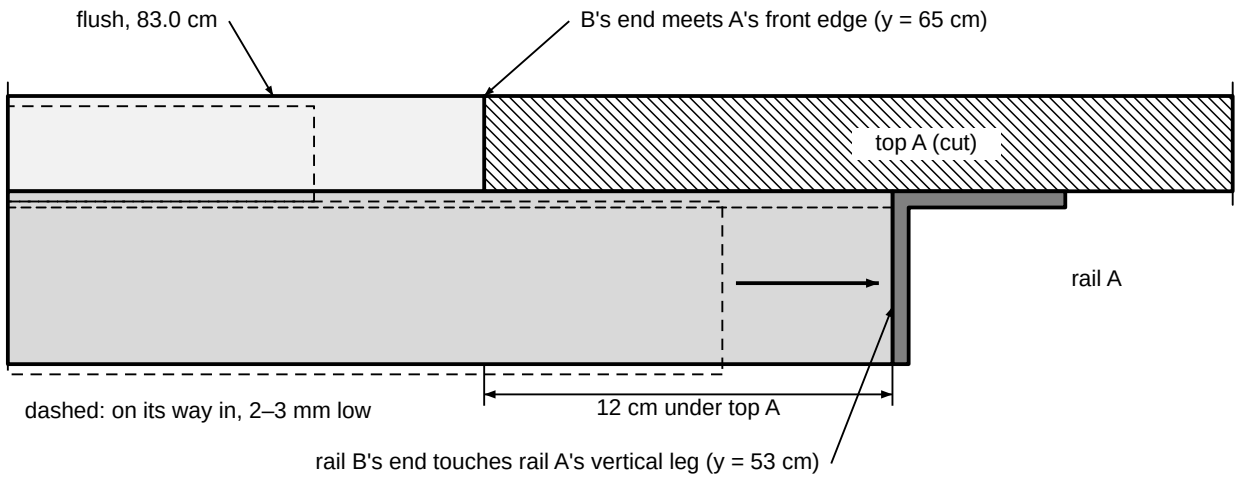


Fig. G-4 · The joint in section, same view as Fig. G-3: top A on the right, y grows to the left (1:2.2). Dashed: top B on its way in, 2–3 mm low. Solid: home. Rail B's flat leg bears on top A's underside for its first 12 cm; its holes 1 and 2 are screwed into top A in chapter H, step 4.

Never force the corner.
If the corner will not meet or come flush, adjust leg height, not the joint. Never hammer, clamp or lever the tops together. If B's end is off along the joint, push ALEX 2 with the pads in the rings; top B moves with it.

Next: chapter H, the joint: plates, rail B's last two screws and the corner fittings.

H · The joint

Both tops stand on their rings (chapter G); top B's end meets top A's front edge and rail B's end touches rail A. You work from below, lying in the knee space. The two gusseted fittings bolted round rail B's end, not the joining plates, carry the inside corner of top B.

SEQUENCE: THE FITTINGS GO ON LAST SOURCE-DEFINED
1 · The two surfaces are flush at the joint, levelled with the feet, never packed (step 1).
2 · Rail B's end touches rail A's vertical leg; rail B's outer face is on the taped 52.0 cm mark; the L is square (step 2).
3 · Joining plates P1 and P2 are screwed across the joint (step 3). They align the tops; they do not carry the corner.
4 · Rail B's screws 1 and 2 are in top A (step 4). The inside fitting sits right under them and would block the driver.
5 · Only then the fittings (steps 5–7), and the bolts tightened last (step 8).

	Part	Qty		Hardware	Qty
P06 **P07**	Rail A, rail B, on the tops (chapter F)	1 + 1	**H03**	#10 × 1 in pan head, Robertson (rail B holes 1, 2)	2
P08	Gusseted 90° strut fitting, 5 mm, legs 100.5 × 41 mm, two 21 × 12 mm slots per leg VERIFIED	2	**H04** **H05**	#10 × 5/8 in pan head, #10 washer (plates)	8 + 8
P11	Mending plate 10 × 3 × 0.12 in (joining plates P1, P2)	2	**H08** **H09** **H10**	3/8-16 × 1½ in hex bolt grade 5; 3/8 flat washer; 3/8-16 nylon-insert nut	6 / 12 / 6

1 **Surfaces flush.** Bring the two surfaces level at the joint by turning the OLOV feet and the round feet, never by packing anything under the joining plates. A fingertip across the joint finds no step.

2 **Check under the corner and the square, while the tops can still move.** Rail B's end touches the face of rail A's vertical leg with no gap, and rail B's outer face lines up with the taped 52.0 cm mark on rail A. The butt joint is closed across its full 65 cm width (no light). If anything is off, push ALEX 2 with the pads in its rings; top B moves with it. Never push or lift top B alone.

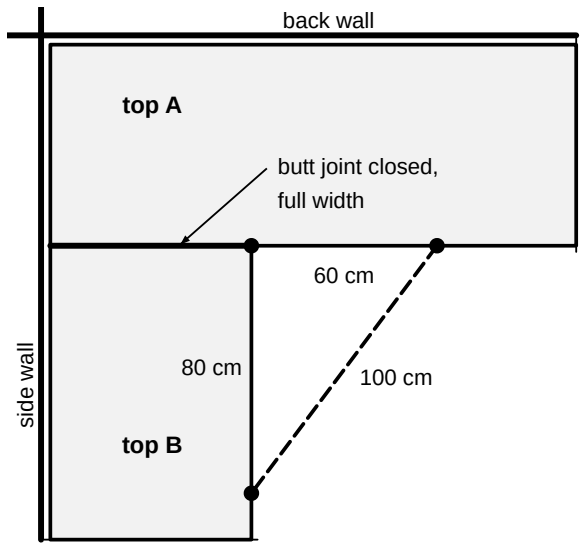


Fig. H-1 · The inside corner seen from above: x grows to the right, y toward you (1:24).

FIELD CHECK, not in the source documents
ASSUMPTION / FIELD CHECK
Optional 3-4-5 triangle: hook the tape on the inside corner and mark 60 cm along A's front edge and 80 cm along B's inside edge; the tape between the two marks reads 100 cm when the L is square. Write down what it reads. If it is off, check the butt joint for a gap first, then push ALEX 2 with the pads in its rings. Never force the corner.

3

Joining plates P1 and P2, from below. Each plate goes on the outline drawn in chapter D: y 53.3–78.7 cm, long axis along y, centred on x = 15 cm (P1) and x = 35 cm (P2). They are not centred on the joint: each starts 3 mm clear of rail A.

1. Hold the plate on its outline and scribe through its four holes.
2. Take the plate off. Pilot each mark 1/8 in, 12 mm deep, with the drill guide's **12 PLATE** puck standing on the wood.
3. Put the plate back and drive H04 #10 × 5/8 in, each with an H05 washer: two screws in each top.

SOURCE-DEFINED

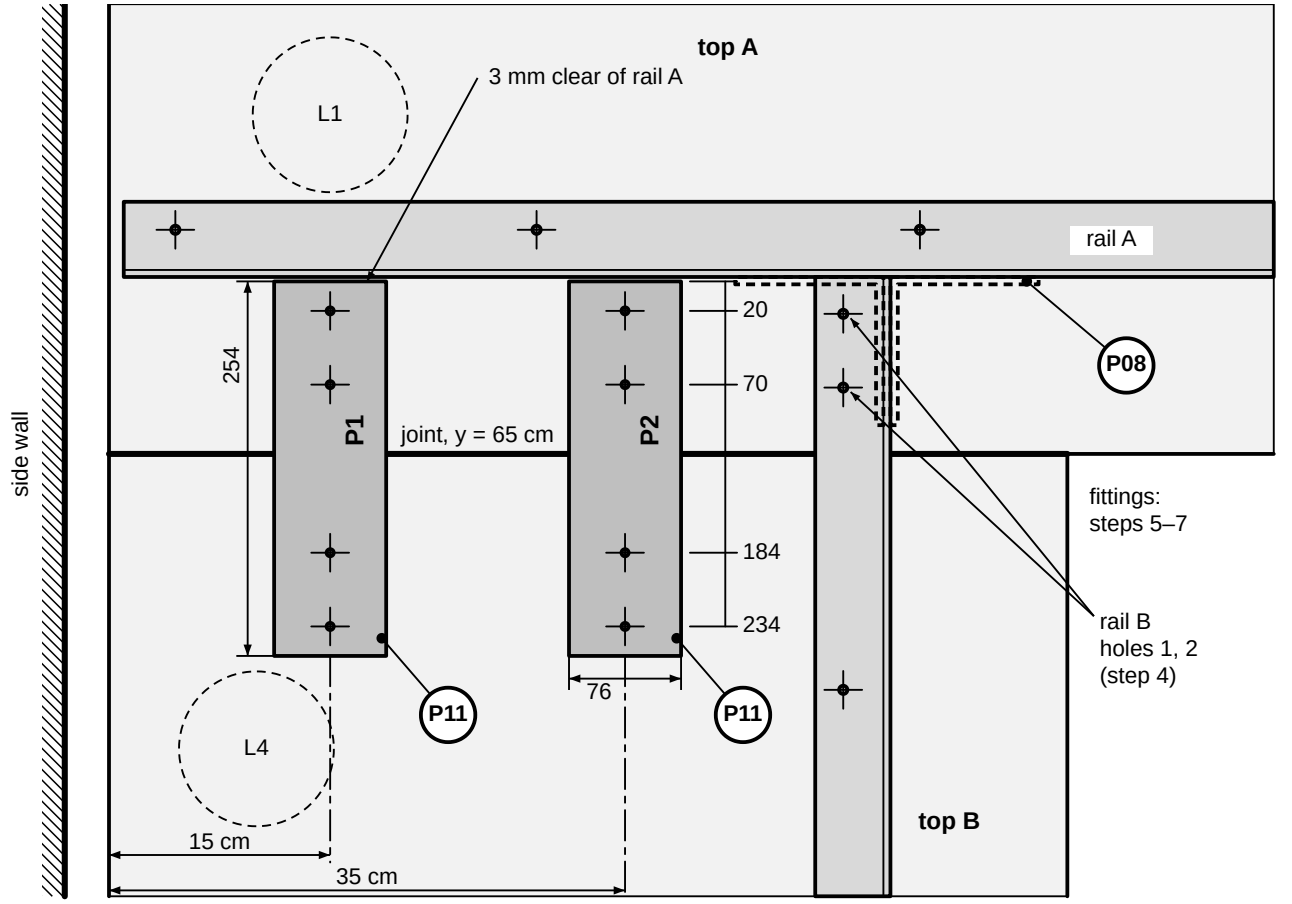
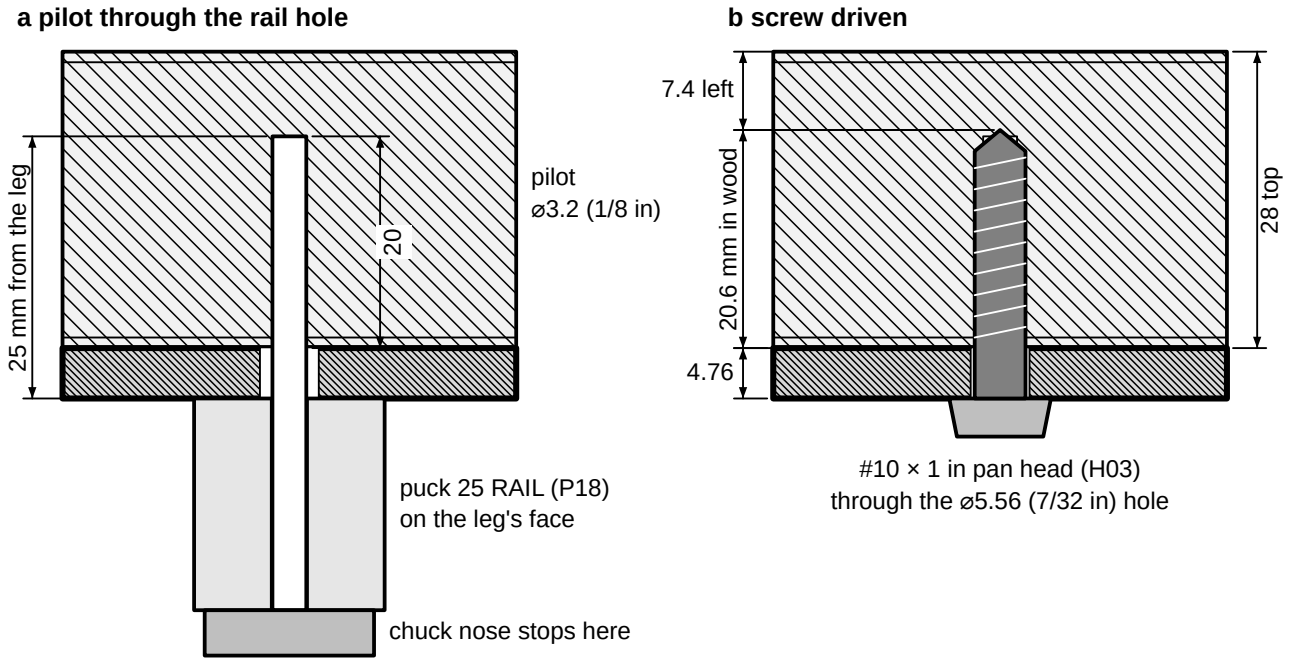


Fig. H-2 · P08 fittings, P11 plates. The corner from above, looking through the tops; x grows to the right, y toward you (1:5). Datum: top A's back-left corner; positions in cm, plate sizes and hole spacings in mm. Plate holes at 20, 70, 184 and 234 mm from the plate end nearest rail A. Dashed: OLOV plates L1 and L4, and the two fittings (steps 5–7).

Plate	Centre x (cm)	Span y (cm)	Screws in top A, y (cm)	Screws in top B, y (cm)	Pilot (mm)
P1	15	53.3–78.7	55.3, 60.3	71.7, 76.7	⌀3.2 (1/8 in) × 12 deep, puck 12 PLATE
P2	35	53.3–78.7	55.3, 60.3	71.7, 76.7	same

Every plate hole is more than 4 cm from an edge, so all four get a screw.

4 **Rail B's screws 1 and 2.** Rail B's first 12 cm lies under top A; its holes 1 and 2 (2.5 cm and 7.5 cm from its corner end) are the only rail holes not yet screwed. Through each, drill a 1/8 in pilot with the drill guide's **25 RAIL** puck standing on the flat leg; it stops the bit 20 mm into the wood. Drive H03 #10 × 1 in with the Robertson #2 bit on the 3 in extension until the head seats. SOURCE-DEFINED



Never more than 21 mm into the 28 mm top. Drawn upright, as for rail B's holes 1 and 2.

Fig. H-3 · One rail screw in section, all mm. The puck stops the pilot 25 mm from the flat leg's face, 20 mm into the wood; the screw passes the 4.76 mm steel and goes 20.6 mm into the 28 mm top, leaving 7.4 mm under the laminate. SOURCE-DEFINED; top thickness MEASURE ON HARDWARE: if a top reads 27 mm nothing changes, this deepest screw still stops 6 mm short of the laminate; under 27 mm, stop and ask.

Every hole below this point is already drilled: the rails' bolt holes were drilled on the bench through a fitting (chapter E). Rail A: 54.9 cm and 59.8 cm (outside fitting), 48.6 cm and 43.7 cm (inside fitting) from the wall end; rail B: 2.9 cm and 7.8 cm from the corner end VERIFIED.

The corner assembly

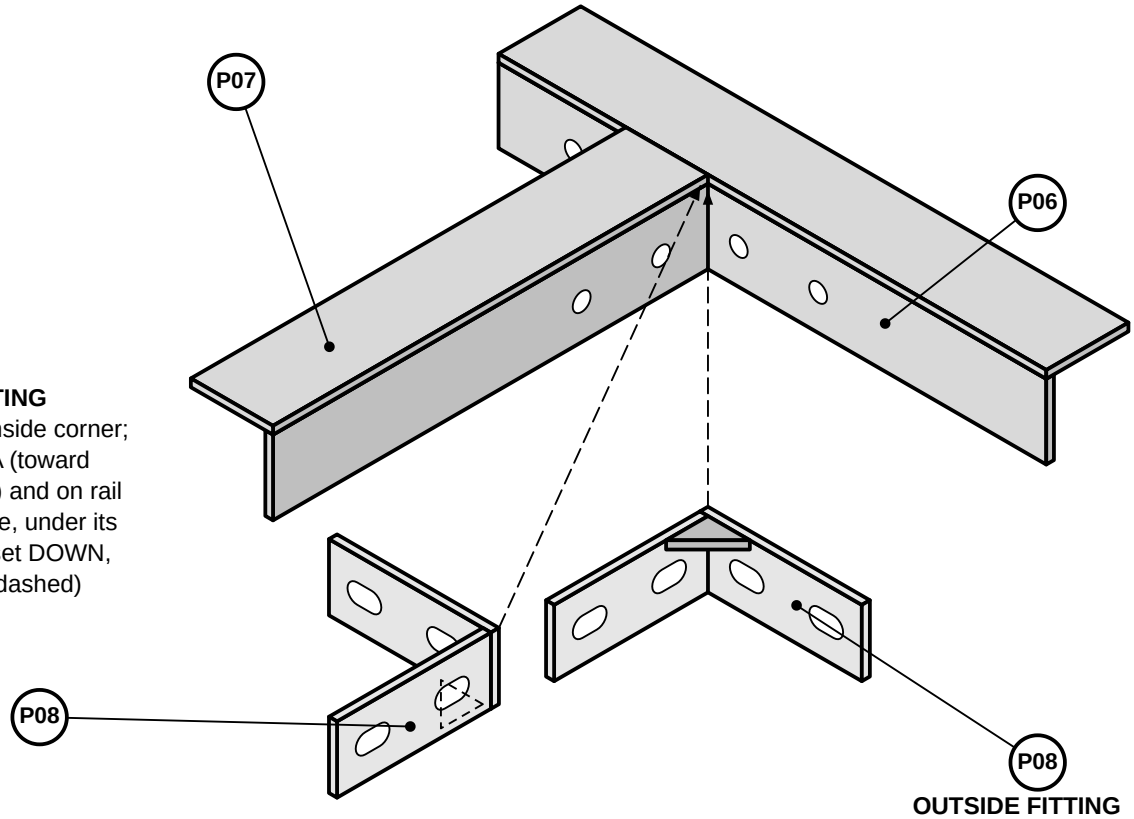


Fig. H-4 · P06 rail A, P07 rail B, P08 fittings. The corner seen from the knee space, from above, with the tops lifted away: rail A's far end is to your right, rail B runs away to your left. Both fittings are drawn pulled out of place; the dashed lines run from each heel to where it goes. They are the same part: the inside one is turned gusset-down, not mirrored. SOURCE-DEFINED placement.

Which bolt goes where

Bolts	Through	Head and washer	Washer and nut
O1, O2	outside fitting + rail A's vertical leg	fitting side, in the knee space	rail A's inner face, under its flat leg
I1, I2	inside fitting + rail A's vertical leg	fitting side, in the pocket under rail B	rail A's inner face, under its flat leg
C1, C2	outside fitting + rail B's vertical leg + inside fitting	outside fitting, knee side	inside fitting, in the pocket

Every bolt goes in **from the fitting side**: four H08 through rail A's leg with one fitting each, two through rail B's leg with a fitting on each side. H09 washer under every head and every nut; H10 nylon-insert nuts.

5 **Outside fitting.** Heel into the outside corner, one leg flat on rail A's vertical leg toward the far end, the other on rail B's outer face, gusset up, facing the knee space. Push O1 and O2 through from the fitting side with a washer under each head; washer and nut on rail A's inner face. Finger-tight.

6 **Inside fitting.** Into the pocket: heel in the inside corner, one leg on rail A's vertical leg toward the wall end, the other on the inner face of rail B's vertical leg, **gusset pointing at the floor**. Gusset-up it would sit 1 mm under the head of rail B's screw 1. I1 and I2 from the fitting side, washers, nuts on rail A's inner face. Finger-tight.

Section at the bolt line, 29 mm below the top, seen from above. Dashed: flat legs above the cut.

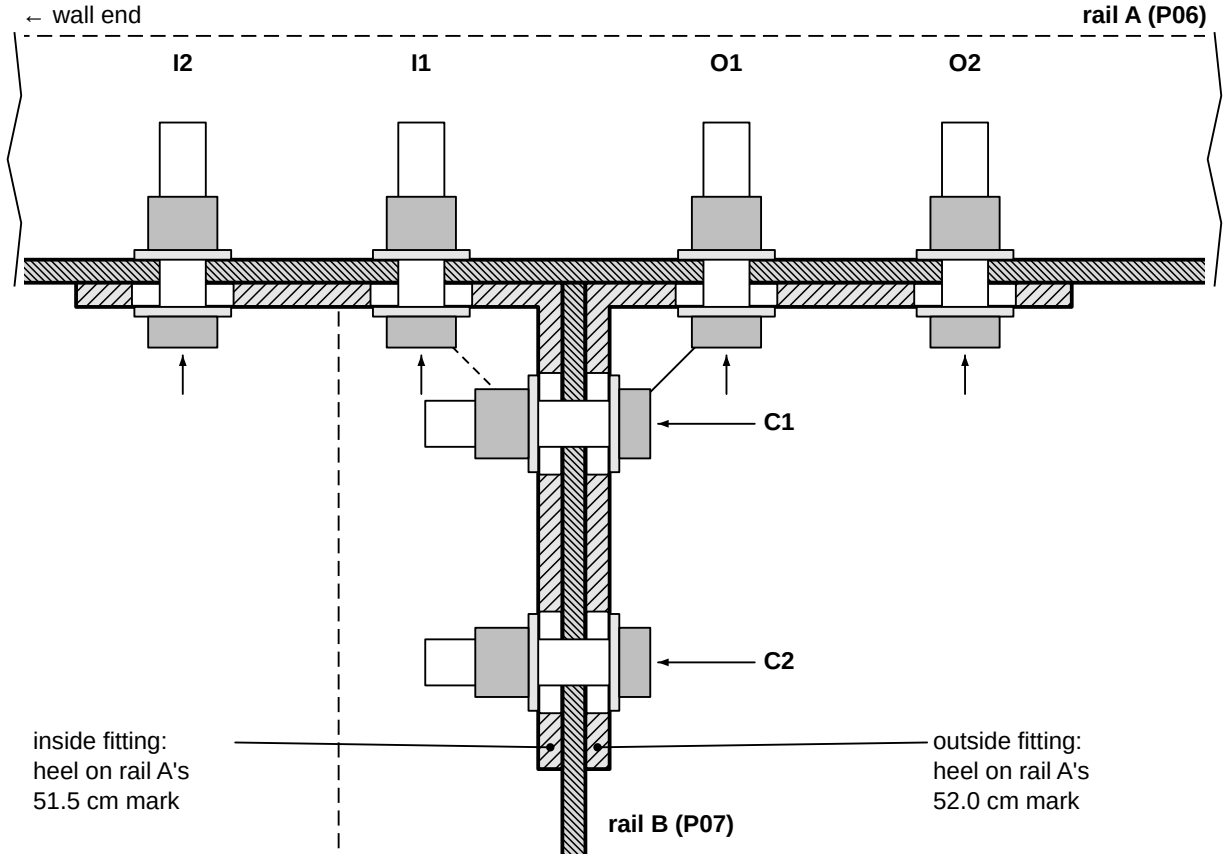
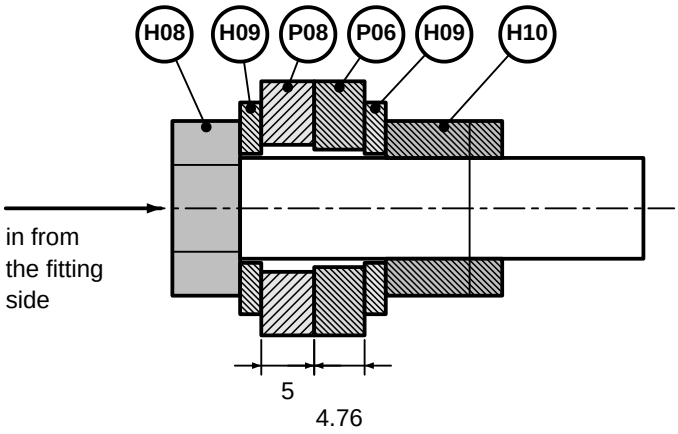


Fig. H-5 · P06 rail A, P07 rail B. Section at the bolt line, seen from above: x grows to the right (wall end on the left), y toward you. The outside fitting's heel is on rail A's 52.0 cm mark, the inside one's on the 51.5 cm mark. Arrows: insertion direction. The cut runs along the fitting slots, so their 21 mm length shows round each bolt.

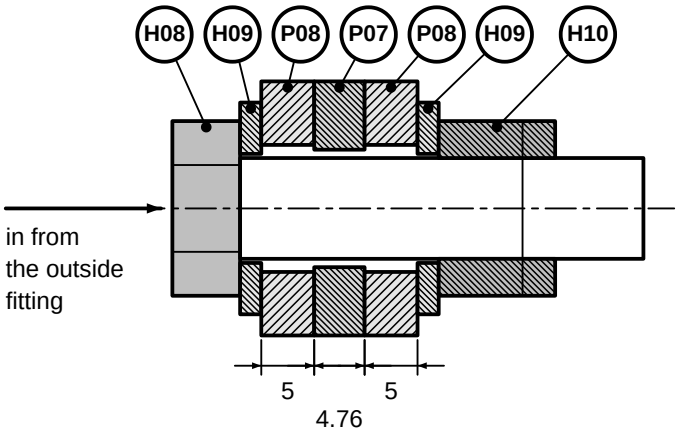
7 **Through bolts.** C1 and C2 from the outside fitting: through the outside fitting, rail B's vertical leg and the inside fitting, washer under the head, washer and nut in the pocket. Finger-tight. All six bolts are now in; none is tight.

Rail A: 4 bolts, one fitting each (O1, O2, I1, I2)



fitting 21 × 12 slot (cut across):
2.5 mm play across, 11.5 mm along
rail hole ⌀11.1 (7/16 in):
1.6 mm of play
so the joint holds by friction:
tighten fully, 25 N·m

Rail B: 2 bolts through outside fitting, rail B and inside fitting (C1, C2)



head and washer on the
outside fitting (knee side);
washer and nut on the
inside fitting, in the pocket
under rail B's flat leg

Fig. H-6 · H08 bolt, H09 washer, H10 nut, P06/P07 rails, P08 fitting. Bolt stack-ups, cut along the bolt, all mm; plates drawn cut off. The fitting's slots are 21 × 12 mm; the cut shows their 12 mm width. VERIFIED fitting; SOURCE-DEFINED bolts.

8 **Check again, then tighten all six.** Fingertip across the joint: no step. Rail B's end still touching rail A. Then tighten every bolt fully, as the panel says.

! **TIGHTENING: 25 N·m (18 ft·lb)** SOURCE-DEFINED
With a torque wrench, **25 N·m (18 ft·lb)**, under the standard dry torque even for a grade 2 bolt. Without one, pull as hard as one hand can on a 15 cm wrench, then stop.
Why tight, not snug: this is a friction joint. The fittings' 21 × 12 mm slots leave 2.5 mm of play across and 11.5 mm along each bolt, the 7/16 in rail holes 1.6 mm, so only the clamping force keeps rail B's corner from settling.
When: only after the two surfaces are flush and rail B's end touches rail A.
Later: if the corner ever drops a millimetre, retighten these six bolts.

9

Quick checks before the mount. If one fails, fix it with the feet or the legs, never at the joint. Heights at the corners are measured in chapter K.

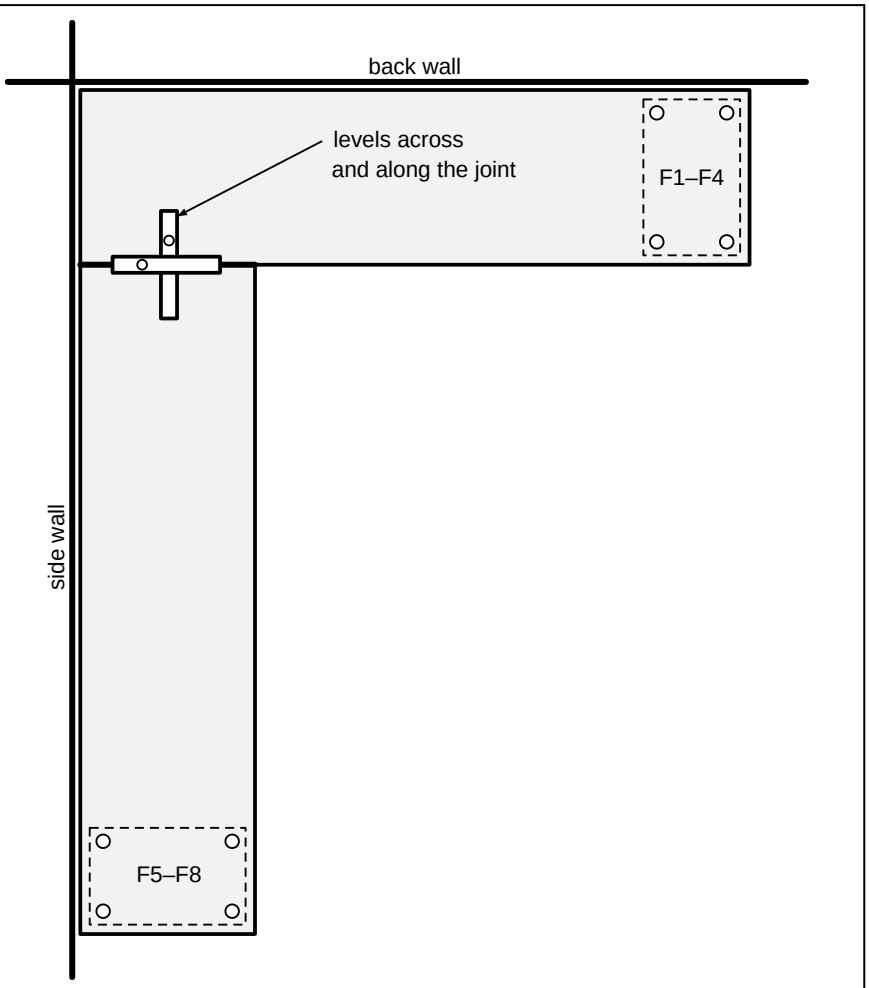


Fig. H-7 · The level positions and pads in plan, seen from above: x grows to the right, y toward you (1:28). Datum: top A's back-left corner.

- **Level:** a level across the joint and along it reads flat both ways; a fingertip finds no step.
- **Pads:** all eight bear. Rock each of F1–F8 sideways by hand; do not turn it, that changes its height. None rocks or lifts.

Load test, sideways push and settling record: chapter K. Never stand or sit on the corner.

Next: chapter I, the monitor mount.

J · Wall-gap trim (optional)

> ! **After chapter I.**
> The clamp pieces wrap round the VIVO clamp backs, so both clamps must already be fitted and tight. The trim goes on before the final push to the walls.

Parts: P19 wall-gap trim (corner, two 80 mm fillers, two clamp pieces, 240 mm segments, the short end pieces, connector tiles), H07 #4 × 1/2 in flat head × 48. Tools: tape measure, drill guide with the **10 EDGE** puck, 5/64 in bit, wall gauge. No figure in this chapter uses part balloons.

1 **Unload the desk.** Take the screens, the PC and everything loose off the desk.

2 **Measure, set, print.** With the desk still against both walls, measure the gap between each top's raw back edge and the wall, in mm, at five or six points along each wall. Take the smallest reading, find it in the table, and set `gap` in `parts/accessories/wall-gap-common.scad` to the value under it. Then print the trim.

MEASURE ON HARDWARE

Smallest gap (mm)	27	28	29	30	31	32	33	under 27 or over 33
Set gap to	26	27	28	29 (as saved)	30	31	32	measure again and ask before printing

Wall	From the wall corner outward: desk positions in cm, piece lengths in mm	Slotted segments (cm)
back	corner (to 20), 80 mm filler (20–28), clamp piece (28–52), 8 × 240 mm, 1 × 50 mm (to 249)	3rd (100–124), 8th (220–244)
side	corner (to 20), 80 mm filler (20–28), clamp piece (28–52), 10 × 240 mm, 1 × 220 mm (to 314)	3rd (100–124)

The corner piece covers each wall's first 20 cm of desk. 24 connector tiles: 11 joints on the back wall, 13 on the side wall. Cut list from parts/README.md.

3 **Pull the desk out, 40 cm from both walls.** Two people, one at each ALEX unit: slide the whole desk away from both walls. The ALEX units slide on their feet and the pads stay in their rings. Nothing comes apart.

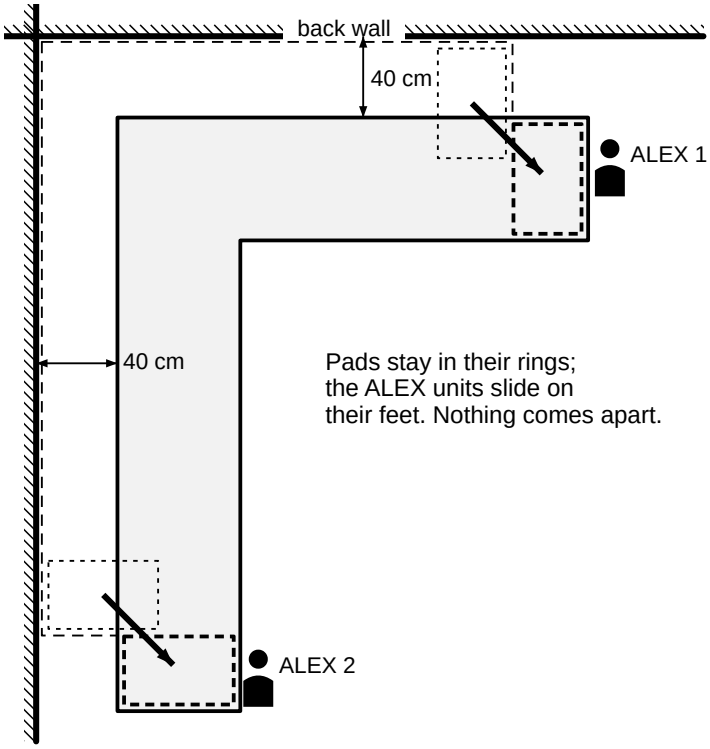
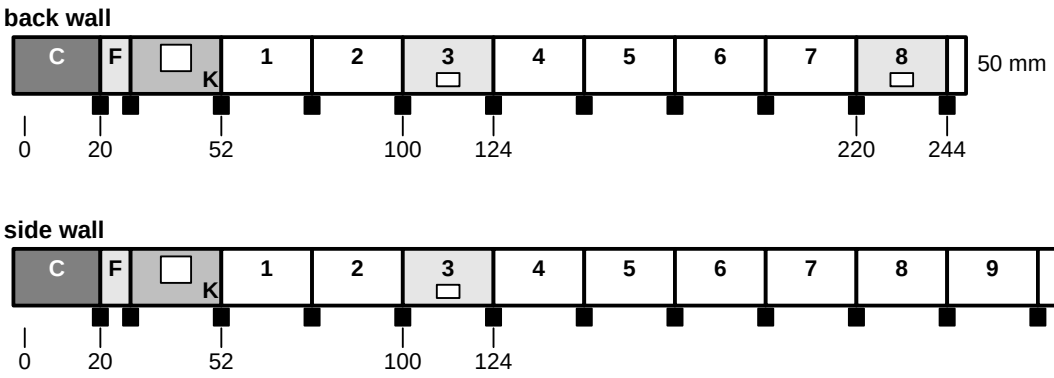


Fig. J-1 · Plan, seen from above: x grows to the right, y toward you, back wall at the top (1:40). Dashed: the desk in place, 3 cm off both walls. Grey: pulled 40 cm off both, so the raw back edges are open to work on.

4 **Build each chain from the corner outward.** Corner piece, 80 mm filler, clamp piece with its notch round the clamp back, then the 240 mm segments (slotted ones where the cut list says), ending with the short piece. Butt every joint and push a connector tile up into the pockets.



Desk positions in cm. C corner · F 80 mm filler · K clamp piece (white: notch round the clamp back)
1, 2 … 240 mm segments, grey with a slot = slotted · ■ connector tile under every joint

Fig. J-2 · Chain order along both walls, straightened out: the wall corner on the left, desk positions in cm growing to the right, piece lengths in mm (1:20). The notch (white, in K) wraps the clamp back, 80 mm long and 17.5 mm proud VERIFIED.

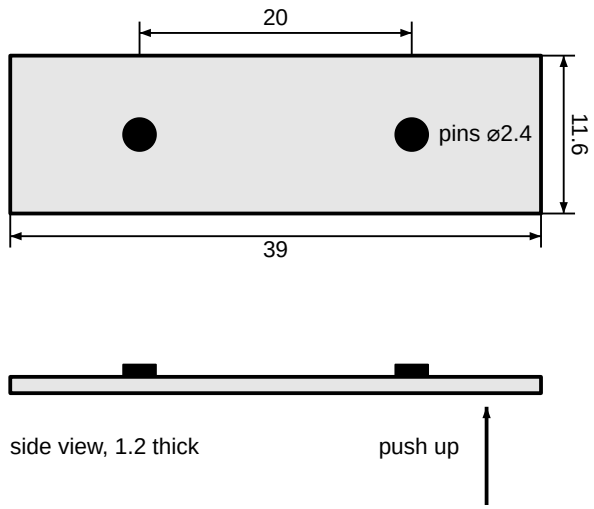


Fig. J-3 · Connector tile (1.8:1, mm), one under every joint, pins up. It bridges the 20 × 12 mm pockets of two butted pieces.

Every piece, the corner included, has a pocket on its underside at each end, over the gap. Butt two pieces, then push a tile up into both pockets until its pins seat: the joint cannot step, bow or drift, and nothing shows on top.

If a wall is out and the notch misses the clamp, change the 80 mm filler's `seg` and reprint it, nothing else.

5 **Pilot and screw.** Through each lip hole, pilot 5/64 in, 10 mm deep, drill horizontal, with the drill guide's **10 EDGE** puck held against the raw edge. Drive an H07 #4 × 1/2 in flat head into each: two per segment, one per corner leg, one each side of a clamp notch.

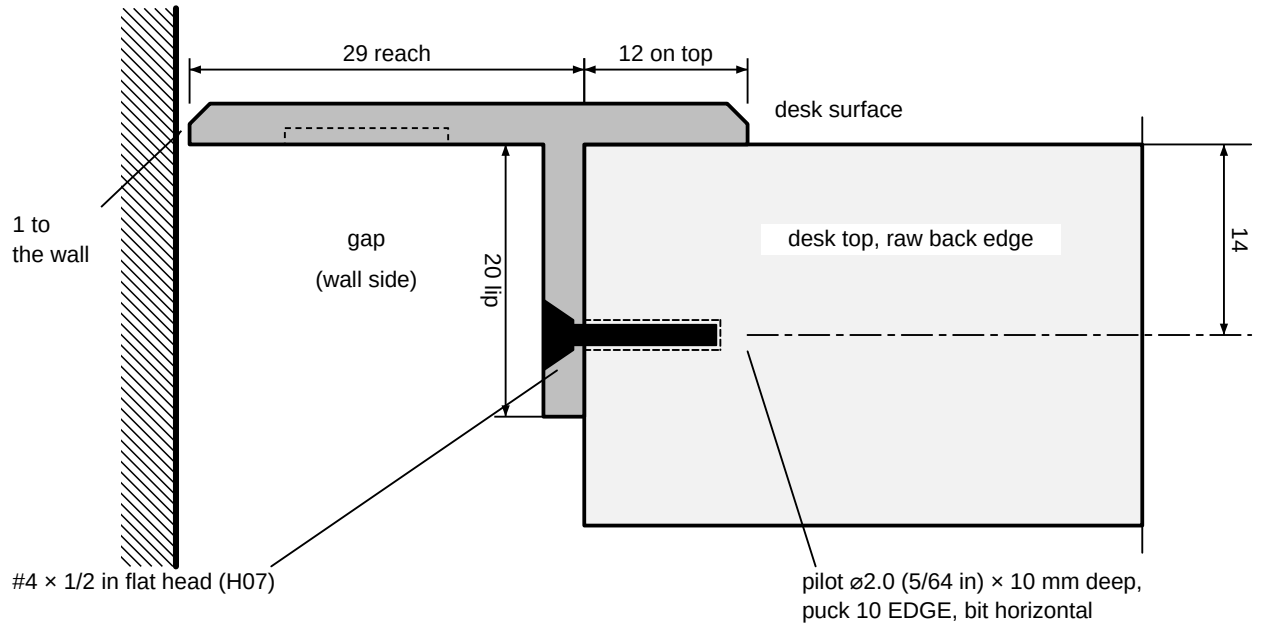


Fig. J-4 · Trim on the raw back edge, section across the edge (1.8:1, mm), seen from the side-wall end looking along the back wall (x grows away from you): the wall on the left, the desk on the right. 12 mm rests on the surface, the 20 mm lip drops into the gap, the screw sits 14 mm below the surface: nothing shows on top and nothing goes into the top face. Profile from parts/accessories/wall-gap-common.scad **SOURCE-DEFINED**; top 28 mm thick **MEASURE ON HARDWARE**.

6 **Slide the desk back.** Two people again, one at each ALEX, until the wall gauge's ALEX 65 leg (6.5 cm) just fits between the wall and each ALEX. The trim's outer edge stops 1 mm short of the wall.

7 **Load the desk again.** Put the screens, the PC and everything else back in their chapter I positions: chapter K checks the desk with everything in place.

Retighten every screw after two weeks.

Next: chapter K, final inspection and load testing.

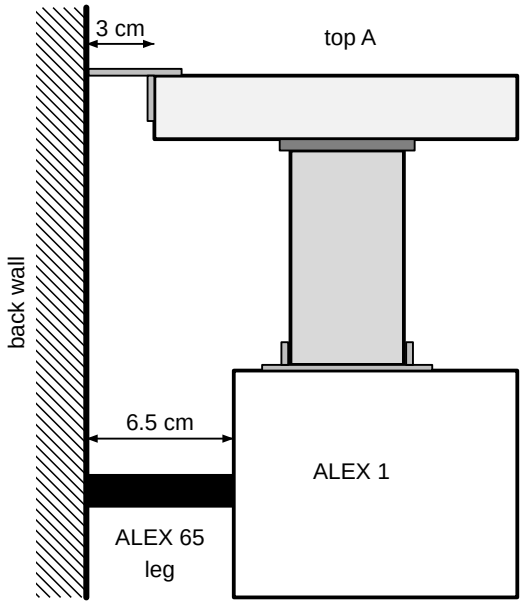


Fig. J-5 · Wall gauge at ALEX 1, section across the back wall, seen from the side-wall end looking along the back wall (x grows away from you): the wall on the left, the room on the right (1:3.3).